

Rishab Vempati

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EDUCATION

Purdue University

West Lafayette, IN

B.S. in Computer Science & Artificial Intelligence, GPA: 3.83/4.00

Aug. 2023 – May 2027

- **Coursework:** Analysis of Algorithms, Data Structures, Operating Systems, Computer Architecture, Database Systems, Introduction to AI, Probability & Statistics, Linear Algebra

EXPERIENCE

Software Engineering Co-op

Aug. 2026 – Dec. 2026

IBM – Watsonx Orchestrate

Austin, TX

- Incoming co-op on the Watsonx Orchestrate team, IBM's AI agent platform for enterprise workflow automation.

Undergraduate Researcher | [Paper under review at UIST '26 \(3rd author\)](#)

Feb. 2026 – Aug. 2026

CIDER Lab, Purdue University – Advisor: Prof. Jason Wu

West Lafayette, IN

- Training uxCUA, a computer-use agent for automated UI usability assessment using RL-based fine-tuning of vision-language models on a dataset of 879 interactive websites with injected usability defects.
- Running large-scale agent rollout generation and evaluation on GPU compute clusters (NVIDIA H100s).
- Developing navigation quality heuristics and reward functions to improve agent exploration behavior.

Software Engineer – Part Time

Aug. 2025 – Feb. 2026

Ekai

Remote

- Built a RAG pipeline using Pinecone to index and retrieve context from 10,000+ docs with 92% retrieval accuracy.
- Deployed the retrieval system as an MCP Server, enabling interoperability across multiple AI agent clients.
- Developed an AI assistant for Microsoft Teams using LLM orchestration for natural language knowledge queries.
- Integrated SQL retrieval and response caching, cutting P95 latency from 8.2s to 1.8s for 50+ concurrent users.

Software Engineering Intern

Jun. 2025 – Aug. 2025

AssetMark

San Francisco, CA

- Migrated legacy .NET/C# communication service to event-driven microservices using Azure Service Bus.
- Implemented dead-letter queues and exponential backoff retry, reducing message failure rate from 0.8% to 0.12%.
- Enabled autoscaling that handled 1M+ quarterly emails with 99.7% delivery success rate during traffic spikes.

Software Engineering Intern

May 2024 – Aug. 2024

Acceleera Payments

San Francisco, CA

- Implemented Kafka streaming pipeline for ISO 20022 payment processing, load-tested to 50K+ daily transactions.
- Containerized 4 microservices with Docker multi-stage builds in GitLab CI/CD, cutting deploy time by 80%.
- Built Grafana dashboards to monitor system health, tracking 850 msg/sec throughput and 120ms P99 latency.

Undergraduate Researcher | [Poster](#)

Jan. 2024 – May 2024

Cisco

West Lafayette, IN

- Developed MLE and Bayesian inference algorithms to select optimal forecasting models for 10K+ product SKUs.
- Built outlier detection and clustering pipelines (Isolation Forest, GMMs), improving forecast accuracy by 3%.

PROJECTS

Parking Lot Detection & Mapping | [Python, PyTorch, OpenCV, Docker](#) | [Code](#)

- Led a 5-person team building a U-Net segmentation model for satellite imagery, achieving 84% IoU accuracy.
- Built automated pipeline ingesting 50GB+ of Bing Maps imagery; contributed 5,000+ annotations to OSM.

YourPingMe – Intelligent Notification Scheduler | [Python, React, Redis, Celery, PostgreSQL](#) | [Code](#)

- Built a unified API wrapper aggregating OpenAI, Anthropic, and Gemini into a single model-switching interface.
- Implemented scheduling engine with Redis/Celery for recurring LLM calls with async SMS/email notifications.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, SQL, JavaScript, TypeScript, Bash, R, HTML/CSS, Assembly

Technologies: FastAPI, Flask, Node.js, .NET, Docker, Kubernetes, Apache Kafka, Redis, CI/CD, Linux, Azure, AWS

AI/ML: PyTorch, TensorFlow, JAX, OpenCV, Pinecone, LangChain, Elasticsearch, LangGraph, MCP, Spark, vLLM

Certifications: AWS Certified Cloud Practitioner, Microsoft Azure Fundamentals (AZ-900)